

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

Nov 2012

EC 208 Analog Electronic Circuits

Date: 07.11.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain series voltage regulator with necessary block diagram and circuit diagram. [05]
- (b) Compare all the power amplifier classes with their efficiency. [02]
- Q - 2 (a) Explain voltage transformation, current transformation, impedance transformation with the help of transformer coupled class A audio power amplifier. [03]
- (b) For an ideal differential amplifier define following. [05]
1. Difference Signal.
 2. Common Mode Signal.
 3. Common Mode Rejection Ratio.
- (c) Define the terms slope error, displacement error, and transmission error. How are they related for an exponential sweep circuit? Derive the relation between them. [06]

OR

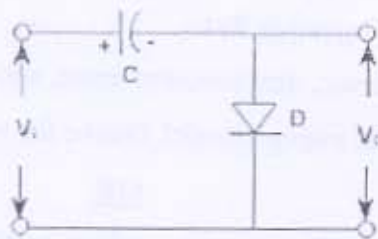
- Q - 2 (a) Find out the value of Transformer turns ratio required to match a 16 ohm speaker load so that the effective load resistance seen at the primary is 10 K Ω . [03]
- (b) Draw the equivalent circuit from which to calculate A_v and A_d for the emitter coupled differential amplifier. [05]
- (c) What do you mean by (a) A voltage time base generator, (b) A current time base generator, and (c) A linear time base generator? And list the applications of time base generators. [06]
- Q - 3 (a) With the help of a neat circuit diagram, explain the working of a simple current sweep. [06]
- (b) What is the relationship between the transfer gain with feedback A_f and that without feedback A ? Define (a) Positive feedback (b) Negative feedback. [04]
- (c) List five characteristics of an amplifier which are modified by negative feedback. [04]

OR

- Q - 3 (a) How the linearity is corrected through adjustment of the driving waveform for a current time base generator? [06]
- (b) Define desensitivity D. for a large value of D, what is A_f ? What is the significance of this result? [04]
- (c) State whether the input resistance R_f is increased or decreased for each feedback amplifier topologies. [04]

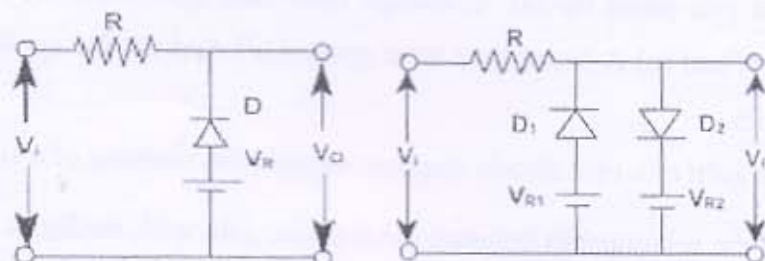
SECTION - II

- Q - 4 (a) Explain the Barkhausen criterion for oscillation. [02]
- (b) Explain the wein bridge oscillator and derive the equation of oscillation frequency for $R_1=R_2=R$ and $C_1=C_2=C$. [05]
- Q - 5 (a) Draw and explain the operation of Emitter coupled clipper. [06]
- (b) State and prove the clamping circuit theorem. [05]
- (c) Draw the Output waveform for the following Circuits. Assume $V_i = 20V(p-p)$ square wave. [03]



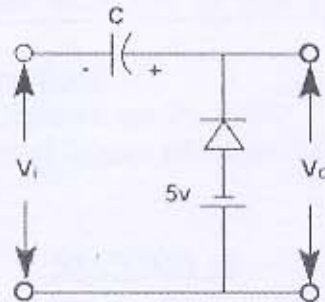
OR

- Q - 5 (a) Draw the Output waveform for the following Circuits. Assume $V_i = 20V(p-p)$ sine wave and $V_R = V_{R1} = V_{R2} = 5V$. [06]



- (b) Explain practical clamping with suitable example. [05]

- (c) Draw the Output waveform for the following Circuits. Assume $V_i = 20V(p-p)$ square wave. [03]



- Q - 6 (a) What do you mean by a current controllable Negative Resistance device and voltage controlled Negative Resistance device? And also give the example of same. [04]
- (b) What do you mean by a negative resistance device? When does a device possess negative resistance? [03]
- (c) How many different output voltages can an 8 bit DAC produce? Explain R-2R Ladder Type D/A convertor. [07]

OR

- Q - 6 (a) What are the applications, advantages and disadvantages of tunnel diode? [04]
- (b) What do you mean by conductivity modulation of UJT? [03]
- (c) What is the advantage of smaller (finer) resolution? Explain Counter Type A/D Converter [07]
